

Linear Algebraic Groups

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Introduction

Notation

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Chapter 1

Basic Theory

§1.1 Motivation

Pell's Equation

§1.2 Categories and Functors

We start with a crash course in category theory.

1.2.1 Definition. A *category* $\mathcal{C}$ consists of the data of

- (i) A class¹ of *objects* $\mathcal{C}$, abusing of notation;
- (ii) For each pair of objects $X, Y \in \mathcal{C}$, a class $\text{Hom}_{\mathcal{C}}(X, Y)$ of *morphisms* from *domain* X to *codomain* Y , denoted $f : X \rightarrow Y$, by which we mean triples² (X, Y, g) where g is any set, called the *graph* of f ;
- (iii) For each triple of object $X, Y, Z \in \mathcal{C}$, a *composition* function

$$\begin{aligned} \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) &\rightarrow \text{Hom}_{\mathcal{C}}(X, Z) \\ (f, g) &\mapsto f \circ g \end{aligned}$$

¹A collection of sets, which is not necessarily a set itself, such as the class of all sets.

²Morally the domain and codomain should be part of the data of a morphism, so that two morphisms are not equal if they do not share their domains and codomains.

such that the following axioms are satisfied

(C1) *Identity*: for each $X \in \mathcal{C}$, there exists an *identity morphism*

$$\text{id}_X : X \rightarrow X$$

such that for any morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow X$,

$$f \circ \text{id}_X = f \quad \text{and} \quad \text{id}_X \circ g = g$$

(C2) *Associativity*: for each triple of morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$, and $h : Z \rightarrow W$, we have

$$(h \circ g) \circ f = h \circ (g \circ f)$$

1.2.2 *Convention*. Write $\text{Hom}(X, Y)$ for $\text{Hom}_{\mathcal{C}}(X, Y)$ when clear.

1.2.3 Exercise. Show that the identity morphism for an object is unique.

1.2.4 Example. In Table 1.1 we provide a list of common categories which we will often reference in the future

Table 1.1: Table of some common categories

Category	Objects	Morphisms
Set	sets	functions
Grp	groups	group maps
Ab	abelian groups	group maps
Ring	commutative rings	ring maps
Alg_R	commutative associative R -algebras	R -algebra maps
Mod_R	R -modules	R -module maps
Mod_K	K -vector spaces	K -linear maps
Vect_K	finite dimensional K -vector spaces	K -linear maps
$\text{Rep}_K(G)$	G -representations over K	G -equivariant maps
Top	topological spaces	continuous functions
PCTop	path-connected topological spaces	continuous functions

where R is a ring, K is a field, and G a finite group.

1.2.5 Definition. Let $f : X \rightarrow Y$ be a morphism in a category, then f is a

- (i) *monomorphism* if $f \circ g = f \circ h$ implies $g = h$ for any $g, h : Z \rightarrow X$,
- (ii) *epimorphism* if $g \circ f = h \circ f$ implies $g = h$ for any $g, h : Y \rightarrow Z$
- (iii) *split monomorphism* if there exists $g : Y \rightarrow X$ such that $g \circ f = \text{id}_X$,
- (iv) *split epimorphism* if there exists $g : Y \rightarrow X$ such that $f \circ g = \text{id}_Y$,
- (v) *isomorphism* if it is a split monomorphism and a split epimorphism,
- (vi) *endomorphism* if $X = Y$,
- (vii) *automorphism* if it is an isomorphism and an endomorphism.

1.2.6 Convention. For an object X in a category $\mathcal{C}$, we will denote by $\text{End}(X) = \text{Hom}(X, X)$ the endomorphisms of X , and we will denote by $\text{Aut}(X)$ the group of automorphisms of X . If there is an isomorphism $f : X \rightarrow Y$ in $\mathcal{C}$, we say X and Y are isomorphic and write $X \cong Y$.

1.2.7 Remark. It's an easy exercise to show split monos (resp. split epis) are monos (resp. epis). Moreover, suppose $\mathcal{C}$ is a category where one can talk about injective and surjective morphisms, in general, split mono (resp. split epi) is a strictly stronger condition than injective (resp. surjective), and injective (resp. surjective) is a strictly stronger condition than mono (resp. epi).

1.2.8 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories, a *covariant functor* (or just simply a *functor*) from $\mathcal{C}$ to $\mathcal{D}$, denoted $F : \mathcal{C} \rightarrow \mathcal{D}$, is the collection of the following data

- (i) for each object $X \in \mathcal{C}$, an object $F(X) \in \mathcal{D}$
- (ii) for each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F[f] : F(X) \rightarrow F(Y)$, such that
 - $F[\text{id}_X] = \text{id}_{F(X)}$ for each $X \in \mathcal{C}$,
 - $F[f \circ g] = F[f] \circ F[g]$ for each $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ in $\mathcal{C}$.

A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$.

1.2.9 Definition. Suppose $\mathcal{C}, \mathcal{D}, \mathcal{E}$ are categories and $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{E}$ are functors, we define their *composite functor* $G \circ F : \mathcal{C} \rightarrow \mathcal{E}$ as the functor that maps $X \mapsto G(F(X))$ for objects and maps the morphisms by $(G \circ F)[f] = G[F[f]]$ for each $f : X \rightarrow Y$.

1.2.10 Definition. Let $\mathcal{C}$ be a category, then the *identity functor* $\text{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ for $\mathcal{C}$ is the functor that maps each object $X \in \mathcal{C}$ by $\text{id}_{\mathcal{C}}(X) = X$ and each morphism $f : X \rightarrow Y$ by $\text{id}_{\mathcal{C}}[f] = f$.

1.2.11 Example. Here are some examples of functors in nature

- (i) the functor $\text{Grp} \rightarrow \text{Set}$ that maps a group to its underlying set and sends morphisms to themselves (functor that “forget” data such as this are called *forgetful functors*)
- (ii) the functor $(-)^{\times} : \text{Ring} \rightarrow \text{Grp}$ which sends a ring R to its multiplicative group of units, and sends a morphism of rings to its restriction on the groups of units,
- (iii) the functor $\text{GL}_n(-) : \text{Ring} \rightarrow \text{Grp}$ which sends a ring R to $\text{GL}_n(R)$ the group of invertible matrices in R , and send a morphism to the obvious

entry-wise group homomorphism.

- (iv) the functor $(-\otimes_R M) : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$ for an R -module M , which sends a R -module N to $N \otimes_R M$ and a morphism $f : N \rightarrow P$ to $f \otimes \text{id}_M$,
- (v) the contravariant functor $(-)^* : \mathbf{Mod}_R^{\text{op}} \rightarrow \mathbf{Mod}_R$ for a ring R which sends an R -module M to its dual module M^* , and a morphism $f^{\text{op}} : M \rightarrow N$ to $f^* : N^* \rightarrow M^*$ by $g \mapsto g \circ f$,
- (vi) the functor $\pi_1(-) : \mathbf{PCTop} \rightarrow \mathbf{Grp}$ where $\mathbf{PCTop}$ is the full subcategory of $\mathbf{Top}$ of path connected spaces, which sends a space X to its fundamental group $\pi_1(X)$ and a continuous map $f : X \rightarrow Y$ to its induced map on fundamental groups $f_* : \pi_1(X) \rightarrow \pi_1(Y)$.

1.2.12 Definition. Define the *opposite category* $\mathcal{C}^{\text{op}}$ of a category $\mathcal{C}$ to be the category with the same objects as $\mathcal{C}$ but with its morphisms $\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathcal{C}}(Y, X)$ for $X, Y \in \mathcal{C}^{\text{op}}$. For each $f : X \rightarrow Y$ in $\mathcal{C}$, we denote by $f^{\text{op}} : Y \rightarrow X$ the corresponding *opposite morphism* in $\mathcal{C}^{\text{op}}$, that is, it has the same mapping as f , but has its domain and codomain swapped.

1.2.13 Definition. A *subcategory* of a category $\mathcal{C}$ is a category $\mathcal{D}$ such that its objects $\mathcal{D} \subseteq \mathcal{C}$ is a subcollection, its morphisms $\text{Hom}_{\mathcal{D}}(X, Y) \subseteq \text{Hom}_{\mathcal{C}}(X, Y)$ for any $X, Y \in \mathcal{D}$, and the composition function of $\mathcal{D}$ is the restriction of that of $\mathcal{C}$. We say $\mathcal{D}$ is a *full subcategory* if further that $\text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y)$ for any $X, Y \in \mathcal{D}$.

1.2.14 Definition. Suppose $\mathcal{C}, \mathcal{D}$ are categories, their product category $\mathcal{C} \times \mathcal{D}$ is the category where objects are pairs (X, Y) where $X \in \mathcal{C}$ and $Y \in \mathcal{D}$, and morphisms are

$$\text{Hom}((X_1, Y_1), (X_2, Y_2)) = \text{Hom}(X_1, X_2) \times \text{Hom}(Y_1, Y_2)$$

with element-wise composition.

1.2.15 Definition. We say that the category $\mathcal{C}$ is *locally small* if $\text{Hom}(X, Y)$ is a set for all X and Y , and *small* if it is locally small and the class of objects of $\mathcal{C}$ is also a set.

1.2.16 Definition. Suppose $\mathcal{C}$ is locally small and $X \in \mathcal{C}$. Define the (*covariant*) *hom-functor* of X

$$\begin{aligned} \text{Hom}(X, -) : \mathcal{C} &\rightarrow \mathbf{Set} \\ Y &\mapsto \text{Hom}(X, Y) \\ (f : Y \rightarrow Z) &\mapsto (f \circ -) \end{aligned}$$

where $(f \circ -) : \text{Hom}(X, Y) \rightarrow \text{Hom}(X, Z)$ maps $g \mapsto f \circ g$. The *contravariant hom-functor* of X is

$$\begin{aligned} \text{Hom}(-, X) : \mathcal{C}^{\text{op}} &\rightarrow \text{Set} \\ Y &\mapsto \text{Hom}(Y, X) \\ (f^{\text{op}} : Z \rightarrow Y) &\mapsto (- \circ f) \end{aligned}$$

where $(- \circ f) : \text{Hom}(Z, X) \rightarrow \text{Hom}(Y, X)$ maps $g \mapsto g \circ f$. The *bivariate hom-functor* of X is

$$\begin{aligned} \text{Hom}(-, -) : \mathcal{C}^{\text{op}} \times \mathcal{C} &\rightarrow \mathcal{D} \\ (X, Y) &\mapsto \text{Hom}(X, Y) \\ (f^{\text{op}}, g) : (X_1, Y_1) \rightarrow (X_2, Y_2) &\mapsto (g \circ - \circ f) \end{aligned}$$

where $(g \circ - \circ f) : \text{Hom}(X_1, Y_1) \rightarrow \text{Hom}(X_2, Y_2)$ maps $h \mapsto g \circ h \circ f$

1.2.17 Remark. Suppose $\mathcal{C}$ is locally small and $X \in \mathcal{C}$. We also use the notations

$$h^X := \text{Hom}(X, -) \quad h_X := \text{Hom}(-, X) \quad h := \text{Hom}(-, -)$$

for the covariant, contravariant, and bivariate hom-functors.

1.2.18 Definition. Let $\mathcal{I}, \mathcal{C}$ be categories, a *diagram* indexed by $\mathcal{I}$ in $\mathcal{C}$ is simply a functor $F : \mathcal{I} \rightarrow \mathcal{C}$. We say that the diagram F *commutes* if for each $X, Y \in \mathcal{I}$, if $f, g \in \text{Hom}(X, Y)$ then $F[f] = F[g]$.

1.2.19 Example. We represent a diagram $F : \mathcal{I} \rightarrow \mathcal{C}$ as a directed multi-graph in the same shape as the index category $\mathcal{I}$, and label the vertices and edges with their images of F in $\mathcal{C}$. For example,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow h & & \downarrow g \\ C & \xrightarrow{e} & D \end{array}$$

and this square commutes if and only if $g \circ f = e \circ h$.

1.2.20 Remark. Functors preserve commutative diagrams. More precisely, let $\mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{C} \rightarrow \mathcal{D}$ a functor. Let $D : \mathcal{I} \rightarrow \mathcal{C}$ be a commutative diagram in $\mathcal{C}$, then $F \circ D : \mathcal{I} \rightarrow \mathcal{D}$ is a commutative diagram in $\mathcal{D}$. This is straightforward from the definition of commutative diagrams.

1.2.21 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories, and $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors, a *natural transformation* from F to G , denoted $\varphi : F \Rightarrow G$, is the data of a morphism $\varphi_X : F(X) \rightarrow G(X)$ for each $X \in \mathcal{C}$ such that the following

diagram commutes

$$\begin{array}{ccc} F(X) & \xrightarrow{\varphi_X} & G(X) \\ F[f] \downarrow & & \downarrow G[f] \\ F(Y) & \xrightarrow{\varphi_Y} & G(Y) \end{array}$$

for all morphisms $f : X \rightarrow Y$ in $\mathcal{C}$. In other words, $G[f] \circ \varphi_X = \varphi_Y \circ F[f]$ for all $f : X \rightarrow Y$ in $\mathcal{C}$. Moreover, if φ_X is an isomorphism for each $X \in \mathcal{C}$, we say φ is a *natural isomorphism*.

1.2.22 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories, and let $E, F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors. Suppose $\varphi : E \Rightarrow F$ and $\psi : F \Rightarrow G$ are natural transformations. We define their (vertical) composition $\psi \circ \varphi : E \Rightarrow G$ as the natural transformation $(\psi \circ \varphi)_X = \psi_X \circ \varphi_X$ for every $X \in \mathcal{C}$.

1.2.23 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{C} \rightarrow \mathcal{D}$ a functor, then define the *identity natural transformation* $\text{id}_F : F \Rightarrow F$ as $(\text{id}_F)_X = \text{id}_{F(X)}$ for each $X \in \mathcal{C}$.

1.2.24 Example. Here are some examples of natural transformations

- (i) let $(-)^{\times}, \text{GL}_n : \text{Ring} \rightarrow \text{Grp}$ be functors defined in [Example 1.2.11](#), then we have the natural transformation $\det : \text{GL}_n(-) \Rightarrow (-)^{\times}$ where $\det_R : \text{GL}_n(R) \rightarrow R^{\times}$ is the determinant,
- (ii) let $(-)^{**} = ((-)^*)^* : \text{Mod}_K \rightarrow \text{Mod}_K$ be the composition of the dual space functor with itself, then there is natural transformation $\text{eval} : \text{id}_{\text{Mod}_K} \Rightarrow (-)^{**}$ given by $\text{eval}_V : V \rightarrow V^{**}$ where $\text{eval}_V(v)(f) = f(v)$ is the evaluation map; it is a natural isomorphism if we replace Mod_K with Vect_K , its full subcategory of finite dimensional vector spaces,
- (iii) let $(-)^* \otimes_R M : \text{Mod}_R^{\text{op}} \rightarrow \text{Mod}_R$ be the composition of $(-)^*$ with $(-) \otimes_R M$ defined in [Example 1.2.11](#), for a ring R and an R -module M ; let $\text{Hom}(-, M) : \text{Mod}_R^{\text{op}} \rightarrow \text{Mod}_R$ be the contravariant hom-functor valued in Mod_R (with the natural module structure inherited from M), there is a natural isomorphism $\varphi : (-)^* \otimes_R M \Rightarrow \text{Hom}(-, M)$ where for R -module N , the map $\varphi_N : N^* \otimes_R M \rightarrow \text{Hom}(N, M)$ is given by $\varphi_N(f \otimes m)(n) = f(n)m$,

1.2.25 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories, the *functor category* from $\mathcal{C}$ to $\mathcal{D}$, denoted $\text{Fun}(\mathcal{C}, \mathcal{D})$, is the category where objects are functors $F : \mathcal{C} \rightarrow \mathcal{D}$, morphisms are natural transformations, and composition is given by (vertical) composition of natural transformations.

1.2.26 Remark. Natural isomorphisms are precisely the isomorphisms in the

functor category.

1.2.27 Definition. Let $\mathcal{C}, \mathcal{D}$ be categories, a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called an *equivalence* if there is a functor $G : \mathcal{D} \rightarrow \mathcal{C}$ such that there are natural isomorphisms $\eta : \text{id}_{\mathcal{C}} \Rightarrow G \circ F$ and $\varepsilon : F \circ G \Rightarrow \text{id}_{\mathcal{D}}$. If there is an equivalence between $\mathcal{C}$ and $\mathcal{D}$, we say they are *equivalent* and write $\mathcal{C} \simeq \mathcal{D}$. Moreover, we call G the *quasi-inverses* of F and call the pair of functors an *equivalence of categories*.

1.2.28 Example. Here are some examples of equivalent categories

- (i) The category $\mathbf{Vect}_K$ of finite dimensional K -vector spaces is equivalent to the category where objects are K^n for $n \geq 0$ and morphisms $\text{Hom}(K^n, K^m)$ are the matrices $M_{n,m}(K)$.
- (ii) The category of intermediate extensions of a finite Galois extension L/K with L -embeddings as morphisms, is equivalent to the category of finite transitive G -sets where $G = \text{Gal}(L/K)$.
- (iii) The category of complex representations of a finite group G (whose morphisms are equivariant maps) is equivalent to the category of $\mathbb{C}[G]$ -modules, and they are also equivalent to $\text{Fun}(BG, \mathbf{Vect}_{\mathbb{C}})$, where BG is the category with a single object $\bullet$, with the endomorphisms of $\bullet$ being group elements, and with the group operation as its composition.
- (iv) R -algebras and ring maps

It is an easy exercise for the reader to spot what the unstated equivalences in the cases above are.

1.2.29 Definition. Suppose $\mathcal{C}, \mathcal{D}$ are categories and $F : \mathcal{C} \rightarrow \mathcal{D}$ is a functor. For each $X, Y \in \mathcal{C}$, then

$$\text{Hom}(X, Y) \xrightarrow{f \mapsto F[f]} \text{Hom}(F(X), F(Y))$$

is a map of sets. We say that F is

- (i) *faithful* if the above map is injective for all $X, Y \in \mathcal{C}$,
- (ii) *full* if the above map is surjective for all $X, Y \in \mathcal{C}$,
- (iii) *fully faithful* if the above map is bijective for all $X, Y \in \mathcal{C}$,

Moreover, call F *essentially surjective* if for all $Y \in \mathcal{D}$ exists $X \in \mathcal{C}$ such that $F(X) \cong Y$ in $\mathcal{D}$.

1.2.30 Theorem. A functor is an equivalence iff it is fully faithful and essentially surjective.

Proof. Technical. See [KS06, Thm. 1.3.13, p. 22]. □

1.2.31 Definition. Let $\mathcal{C}$ be a locally small category. Define

$$\widehat{\mathcal{C}} := \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set}) \quad \check{\mathcal{C}} := \text{Fun}(\mathcal{C}, \text{Set})$$

as the *category of presheaves* of $\mathcal{C}$ and *category of copresheaves* of $\mathcal{C}$ respectively. The *Yoneda embedding* is the contravariant functor

$$\begin{aligned} h^\bullet : \mathcal{C}^{\text{op}} &\rightarrow \check{\mathcal{C}} \\ X &\mapsto h^X = \text{Hom}(X, -) \\ (f^{\text{op}} : Z \rightarrow Y) &\mapsto [X \mapsto (- \circ f)] \end{aligned}$$

where $(- \circ f) : \text{Hom}(Z, X) \rightarrow \text{Hom}(Y, X)$ is as in Definition 1.2.16. Dually, the *co-Yoneda embedding* is

$$\begin{aligned} h_\bullet : \mathcal{C} &\rightarrow \widehat{\mathcal{C}} \\ X &\mapsto h_X = \text{Hom}(-, X) \\ (f : Y \rightarrow Z) &\mapsto [X \mapsto (f \circ -)] \end{aligned}$$

where $(f \circ -) : \text{Hom}(X, Y) \rightarrow \text{Hom}(X, Z)$ is as in Definition 1.2.16.

1.2.32 Lemma (Yoneda). Let $\mathcal{C}$ be a locally small category. Suppose $X \in \mathcal{C}$ and $F : \mathcal{C} \rightarrow \text{Set}$ is a functor, then there are bijections Φ and Ψ , shown below,

$$\begin{array}{ccc} & \Phi & \\ \text{Hom}_{\check{\mathcal{C}}}(h^X, F) & \xrightarrow{\quad} & F(X) \\ & \Psi & \end{array}$$

which are inverses of each other.

Proof. For a natural transformation $\varphi : h^X \Rightarrow F$ and a morphism $f : X \rightarrow Y$ in $\mathcal{C}$, the diagram

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(X, X) & \xrightarrow{\varphi_X} & F(X) \\ (f \circ -) \downarrow & & \downarrow F[f] \\ \text{Hom}_{\mathcal{C}}(X, Y) & \xrightarrow{\varphi_Y} & F(Y) \end{array}$$

commutes. Thus $\varphi_Y \circ (f \circ -) = F[f] \circ \varphi_X$. Evaluating at id_X on both sides yields the identity

$$\varphi_Y(f) = F[f](\varphi_X(\text{id}_X))$$

therefore the natural transformation φ is completely determined by $u = \varphi_X(\text{id}_X)$. Therefore, let

$$\Phi : \text{Hom}_{\mathcal{C}}(h^X, F) \rightarrow F(X) \quad \Phi(\varphi) = \varphi_X(\text{id}_X)$$

and, by the same identity, the inverse of Φ is naturally

$$\Psi : F(X) \rightarrow \text{Hom}_{\mathcal{C}}(h^X, F) \quad \Psi(u) = \varphi$$

where $\varphi_Y : \text{Hom}(X, Y) \rightarrow F(Y)$ is given by $\varphi_Y(f) = (F[f])(u)$, for each $Y \in \mathcal{C}$. It is not hard for the reader to verify that indeed Ψ and Φ are inverses of each other. $\square$

1.2.33 Remark. The preceding [Lemma 1.2.32](#) is known as *Yoneda lemma* or the *fundamental theorem of category theory*. Notice that the definition of Φ and Ψ does not involve in X and F , therefore one can upgrade the lemma to say that there is a mutually inverse pair of natural isomorphism

$$\begin{array}{ccc} & \Phi & \\ \text{Hom}_{\mathcal{C}}(h^\bullet, -) & \xrightleftharpoons{\quad} & (-)(\bullet) \\ & \Psi & \end{array}$$

where the left hand side functor $\text{Hom}_{\mathcal{C}}(h^\bullet, -) : \mathcal{C} \times \check{\mathcal{C}} \rightarrow \mathbf{Set}$ is given by the composition

$$\mathcal{C} \times \check{\mathcal{C}} \xrightarrow{h^\bullet \times \text{id}_{\check{\mathcal{C}}}} \check{\mathcal{C}} \times \check{\mathcal{C}} \xrightarrow{\text{Hom}_{\check{\mathcal{C}}}(-, -)} \mathbf{Set}$$

and the right hand side functor, called the *evaluation functor*, is given by

$$(-)(\bullet) : \mathcal{C} \times \check{\mathcal{C}} \rightarrow \mathbf{Set} \quad (X, F) \mapsto F(X)$$

and sends a morphism $(f : X \rightarrow Y, \varphi : F \Rightarrow G)$ to the morphism $\varphi_Y \circ F[f] = G[f] \circ \varphi_X$. Same constructions for Φ and Ψ will work, hence we have shown that the Yoneda lemma is *functorial* or *natural* in the variables $X \in \mathcal{C}$ and $F \in \check{\mathcal{C}}$.

1.2.34 Exercise. Prove a dual and functorial version of the Yoneda lemma: let $\mathcal{C}$ be a locally small category. Suppose $X \in \mathcal{C}$ and $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$, then construct natural isomorphisms

$$\begin{array}{ccc} & \Pi & \\ \text{Hom}_{\mathcal{C}}(h_\bullet, -) & \xrightleftharpoons{\quad} & (-)(\bullet) \\ & \Theta & \end{array}$$

which are inverses of each other, where you should define the functor

$\text{Hom}_{\mathcal{C}}(h_{\bullet}, -) : \mathcal{C} \times \widehat{\mathcal{C}} \rightarrow \mathbf{Set}$ and the *coevaluation functor* $(-)(\bullet) : \mathcal{C} \times \widehat{\mathcal{C}} \rightarrow \mathbf{Set}$ dually to those in [Remark 1.2.33](#).

1.2.35 Theorem. *Let $\mathcal{C}$ be a locally small category, then the Yoneda and co-Yoneda embeddings*

$$h^{\bullet} : \mathcal{C}^{\text{op}} \rightarrow \check{\mathcal{C}} \quad \text{and} \quad h_{\bullet} : \mathcal{C} \rightarrow \widehat{\mathcal{C}}$$

are fully faithful.

Proof. We will only show $h^{\bullet}$ is fully faithful: for the dual $h_{\bullet}$, the proof is completely symmetric. Let $X, Y \in \mathcal{C}$. By the Yoneda lemma [Lemma 1.2.32](#), there is a bijection

$$\Phi : \text{Hom}_{\check{\mathcal{C}}}(h^X, h^Y) \rightarrow h^Y(X) = \text{Hom}_{\mathcal{C}}(Y, X)$$

by specifying $F = h^Y$. One can verify that this gives the inverse for the map

$$\text{Hom}_{\mathcal{C}}(Y, X) = \text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) \xrightarrow{f \mapsto h^{\bullet}[f]} \text{Hom}_{\check{\mathcal{C}}}(h^X, h^Y)$$

Thus $h^{\bullet}$ is fully faithful. $\square$

1.2.36 Lemma. *Let $\mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{C} \rightarrow \mathcal{D}$ a fully faithful functor, then the morphism $F[f] : F(X) \rightarrow F(Y)$ is an isomorphism in $\mathcal{D}$ if and only if $f : X \rightarrow Y$ is an isomorphism in $\mathcal{C}$.*

Proof. Assume $f : X \rightarrow Y$ is an isomorphism in $\mathcal{C}$, then there exists $g : Y \rightarrow X$ with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. Applying functor F , we get $F[g] \circ F[f] = \text{id}_{F(X)}$ and $F[f] \circ F[g] = \text{id}_{F(Y)}$, hence $F[f]$ has inverse $F[g]$, which makes it an isomorphism in $\mathcal{D}$. Conversely, by fullness of F , choose $g : Y \rightarrow X$ with $F[g] = F[f]^{-1}$. Then we have $F[f \circ g] = F[f] \circ F[g] = \text{id}_{F(X)}$ so by faithfulness of F , we have $f \circ g = \text{id}_X$. By symmetric arguments, we have $g \circ f = \text{id}_Y$, so g is the inverse of f . Therefore f is an isomorphism in $\mathcal{C}$. $\square$

1.2.37 Theorem. *Let $\mathcal{C}$ be a locally small category, then for a morphism $f : X \rightarrow Y$ in $\mathcal{C}$, TFAE*

- (i) $f : X \rightarrow Y$ is an isomorphism in $\mathcal{C}$,
- (ii) $h_{\bullet}[f] : h_X \rightarrow h_Y$ is an isomorphism in $\widehat{\mathcal{C}}$,
- (iii) $h^{\bullet}[f^{\text{op}}] : h^Y \rightarrow h^X$ is an isomorphism in $\check{\mathcal{C}}$,

Proof. Straightforward by [Theorem 1.2.35](#) and [Lemma 1.2.36](#). $\square$

1.2.38 Remark. What [Theorem 1.2.35](#) and [Theorem 1.2.37](#) tells us is that some presheaves (resp. copresheaves) on a category can be represented as ob-

jects in the category, namely these are the ones in the image of the Yoneda (resp. co-Yoneda) embedding.

1.2.39 Definition. Let $\mathcal{C}$ be a locally small category, we call a presheaf $F \in \check{\mathcal{C}}$ (resp. copresheaf $F \in \hat{\mathcal{C}}$) *representable* by an object $X \in \mathcal{C}$ if $h^X \cong F$ (resp. $h_X \cong F$).

§1.3 Schemes and Group Schemes

Unless specified otherwise, fix a ring R , which we shall call our *ground ring*.

1.3.1 Convention. We assume rings are commutative and unital, and algebras are associative, commutative, and unital. In particular, in our convention an algebra A over a ring R is equivalent to the data of a ring A with a ring map $\varphi : R \rightarrow A$, viewing the scalar multiplication as $r \cdot a = \varphi(r)a$. We will use this equivalence frequently and often implicitly.

1.3.2 Definition. A R -*functor* is a functor

$$X : \mathbf{Alg}_R \rightarrow \mathbf{Set}$$

and the *category of R -functors* is the functor category $\mathbf{Fun}_R := \mathbf{Fun}(\mathbf{Alg}_R, \mathbf{Set})$.

1.3.3 Definition. Suppose A is a R -algebra, its *spectrum* is the R -functor

$$\mathrm{Spec}(A) := \mathrm{Hom}_{\mathbf{Alg}_R}(A, -) : \mathbf{Alg}_R \rightarrow \mathbf{Set}$$

An *affine R -scheme* is the spectrum of a R -algebra. The *affine n -space* is the affine R -scheme

$$\mathbb{A}_R^n := \mathrm{Spec}(R[x_1, \dots, x_n])$$

The category $\mathbf{Aff}_R$ of affine R -schemes is the full subcategory of all affine R -schemes of $\mathbf{Fun}_R$.

1.3.4 Remark. The notion of affine R -schemes generalizes the notion of affine varieties. Readers who have experience in algebraic geometry might have been taught that an affine K -variety for some algebraically closed field K is a subset of K^n for some n given by the vanishing set of some polynomials $f_1, \dots, f_m \in K[x_1, \dots, x_n]$, that is, the set

$$\{(x_1, \dots, x_n) \in K^n : f_1(x_1, \dots, x_n) = \dots = f_m(x_1, \dots, x_n) = 0\}$$

For our purposes, however, we take a *functorial* perspective to varieties. To see what this is about, we first make the important observation that the

above set can be naturally identified as the set

$$X(K) = \text{Hom}_{\text{Alg}_K} \left(\frac{K[x_1, \dots, x_n]}{(f_1, \dots, f_m)}, K \right) \quad \text{where} \quad X = \text{Spec} \left(\frac{K[x_1, \dots, x_n]}{(f_1, \dots, f_m)} \right)$$

Namely, each $\phi \in X(K)$ can be identified with $(\phi(x_1), \dots, \phi(x_n)) \in K^n$. Similarly, for polynomials $f_1, \dots, f_m \in R[x_1, \dots, x_n]$, the set

$$X(S) = \text{Hom}_{\text{Alg}_R} \left(\frac{R[x_1, \dots, x_n]}{(f_1, \dots, f_m)}, S \right) \quad \text{where} \quad X = \text{Spec} \left(\frac{R[x_1, \dots, x_n]}{(f_1, \dots, f_m)} \right)$$

for an R -algebra S can be identified as the set

$$\{(x_1, \dots, x_n) \in S^n : f_1(x_1, \dots, x_n) = \dots = f_m(x_1, \dots, x_n) = 0\}$$

where $f_1, \dots, f_m$ are identified as their images along the natural map $R[x_1, \dots, x_n] \rightarrow S[x_1, \dots, x_n]$ induced by the structure map $R \rightarrow S$. Thus, what the spectrum of a R -algebra encodes is the vanishing sets of a set of polynomials over each R -algebra.

1.3.5 Definition. Suppose that $X = \text{Spec}(A)$ is an affine R -scheme where A is a R -algebra. Let S be another R -algebra then we call the set

$$X(S) = \text{Hom}_{\text{Alg}_R}(A, S)$$

the S -points of X . When S is a field, it is also called the S -rational points of X .

1.3.6 Remark. Recall that a R -algebra A is said to be finitely generated if there is a surjective map $R[x_1, \dots, x_n] \rightarrow A$, or equivalently $A = R[x_1, \dots, x_n]/I$ for some ideal $I \subseteq R[x_1, \dots, x_n]$. If R is noetherian, that is, each of its ideals is finitely generated, then by Hilbert's basis theorem, so is $R[x_1, \dots, x_n]$, whence the finitely generated R -algebras are of the form $R[x_1, \dots, x_n]/(f_1, \dots, f_m)$.

1.3.7 Remark. One huge advantage that affine schemes provide is their treatment of *nilpotence*. The vanishing set of a polynomial $f \in K[x_1, \dots, x_n]$ for K a field is the same as that of f^2 . However,

$$\text{Spec}(K[x_1, \dots, x_n]/(f)) \neq \text{Spec}(K[x_1, \dots, x_n]/(f^2))$$

with the right hand side being thought of as having an *infinitesimal thickening* or *nilpotent thickening*.

1.3.8 Remark. Let S be a R -algebra, then $\mathbb{A}_R^1(S) = \text{Hom}_{\text{Alg}_R}(R[x], S)$ is

canonically a R -algebra by

$$\begin{aligned}(\phi + \psi)(f) &= \phi(f) + \psi(f) \\ (\phi\psi)(f) &= \phi(f)\psi(f) \\ (r\phi)(f) &= r\phi(f)\end{aligned}$$

for each $\phi, \psi \in \mathbb{A}_R^1(S)$ and $r \in R$, where $f \in R[x]$. Note that naturally $\mathbb{A}_R^1(S) \cong S$ as R -algebras.

1.3.9 Definition. For an affine R -scheme $X : \mathbf{Alg}_R \rightarrow \mathbf{Set}$, its *coordinate ring* (resp. *coordinate algebra*) is the ring (resp. R -algebra)

$$O(X) := \mathrm{Hom}_{\mathbf{Aff}_R}(X, \mathbb{A}_R^1)$$

with the algebraic operations given as follows: given $f, g \in O(X)$ and $r \in R$

$$\begin{aligned}(f + g)_S(\phi) &= f_S(\phi) + g_S(\phi) \\ (fg)_S(\phi) &= f_S(\phi)g_S(\phi) \\ (rf)_S(\phi) &= rf_S(\phi)\end{aligned}$$

where $\phi \in X(S)$, for each R -algebra S . Moreover, define the contravariant functor

$$O : \mathbf{Aff}_R^{\mathrm{op}} \rightarrow \mathbf{Alg}_R \quad O = \mathrm{Hom}_{\mathbf{Aff}_R}(-, \mathbb{A}_R^1)$$

where for a morphism $\varphi : X \rightarrow Y$ of affine R -schemes, the induced map

$$O[\varphi^{\mathrm{op}}] = \varphi^{\#} : O(Y) \rightarrow O(X) \quad \psi \mapsto \psi \circ \varphi$$

is called the *induced regular map* of φ .

1.3.10 Definition. Define the contravariant functor

$$\mathrm{Spec} : \mathbf{Alg}_R^{\mathrm{op}} \rightarrow \mathbf{Aff}_R \quad A \mapsto \mathrm{Spec}(A)$$

where for each morphism $\varphi : A \rightarrow B$ of R -algebras, there is the induced natural transformation

$$\varphi^* : \mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$$

called the *pullback* given by

$$\varphi_S^* : \mathrm{Spec}(B)(S) \rightarrow \mathrm{Spec}(A)(S) \quad \psi \mapsto \psi \circ \varphi$$

for each R -algebras S .

1.3.11 Theorem. *The pair of functors Spec and O is an equivalence of*

categories

$$\begin{array}{ccc} & \text{Spec} & \\ \text{Alg}_R^{\text{op}} & \xrightarrow{\quad} & \text{Aff}_R \\ & \xleftarrow{\quad O} & \end{array}$$

Proof. Suppose A is a R -algebra, then applying [Lemma 1.2.32](#), namely, the Yoneda lemma

$$O(\text{Spec}(A)) = \text{Hom}_{\text{Aff}_R}(h^A, \mathbb{A}_R^1) \cong \mathbb{A}_R^1(A) \cong A$$

where we leave it to the reader to check that the relevant bijection is an isomorphism of K -algebras. Conversely, let $X = \text{Spec}(A)$, then there is an induced natural isomorphism of hom-functors

$$\text{Spec}(O(X)) = \text{Hom}_{\text{Alg}_R}(O(\text{Spec}(A)), -) \cong \text{Hom}_{\text{Alg}_R}(A, -) = X$$

by [Theorem 1.2.37](#). Therefore Spec and O are quasi-inverses of each other. $\square$

1.3.12 Definition. A R -group functor is a functor

$$G : \text{Alg}_R \rightarrow \text{Grp}$$

The category of R -group functors is the functor category $\text{GFun}_R = \text{Fun}(\text{Alg}_R, \text{Grp})$.

1.3.13 Definition. An affine R -group scheme is a R -group functor $G : \text{Alg}_R \rightarrow \text{Grp}$ such that its composition with the forgetful functor to Set

$$\tilde{G} : \text{Alg}_R \xrightarrow{G} \text{Grp} \rightarrow \text{Set}$$

is isomorphic to an affine R -scheme. Further, we say an affine R -group scheme G is an (affine) algebraic group over R if $\tilde{G}$ is of finite type, i.e. it is the spectrum of a finitely generated R -algebra. Define GAff_R and GAlg_R , the categories of affine R -group schemes and (affine) algebraic groups over R , as full subcategories of GFun_R respectively.

1.3.14 Convention. In this text, “algebraic group” always means affine algebraic group. However, it is important to keep in mind that there are things like abelian varieties which are considered algebraic groups in broader context but are by no means affine.

1.3.15 Example. Here are some examples of algebraic groups over R ,

(i) the additive group

$$\mathbb{G}_a : \text{Alg}_R \rightarrow \text{Grp} \quad S \mapsto (S, +)$$

where $\widetilde{\mathbb{G}}_a \cong \mathbb{A}_R^1$,

- (ii) the *multiplicative group*, also known as the 1-torus,

$$\mathbb{G}_m : \mathbf{Alg}_R \rightarrow \mathbf{Grp} \quad S \mapsto S^\times$$

where $\widetilde{\mathbb{G}}_m \cong \mathrm{Spec}(R[x, y]/(xy - 1))$,

- (iii) the *multiplicative group of n -th roots of unity*,

$$\mu_n : \mathbf{Alg}_R \rightarrow \mathbf{Grp} \quad S \mapsto \{a \in S^\times : a^n = 1\}$$

where $\widetilde{\mu}_n \cong \mathrm{Spec}(R[x]/(x^n - 1))$,

1.3.16 Definition. Let S be an R -algebra.

- (i) If M is an R -module (resp. R -algebra), define the *base change* (or *extension of scalars*) of M from R to S as the S -module (resp. S -algebra) $M_S = M \otimes_R S$. Define the *base change functor* $(-)_S = (-) \otimes_R S : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$ (resp. $(-)_S = (-) \otimes_R S : \mathbf{Alg}_R \rightarrow \mathbf{Alg}_S$).
- (ii) If M is a S -module (resp. R -algebra), define its *Weil restriction* (or *restriction of scalars*) from S to R as the R -module (resp. R -algebra) $M_R = \mathrm{Res}_{S/R} M$ with the same underlying group (resp. ring) as M with scalar multiplication given by $rm = \varphi(r)m$ for $r \in R$ and $m \in M$ where $\varphi : R \rightarrow S$ is the structure map of S . Define the *Weil restriction functor* $(-)_R = \mathrm{Res}_{S/R} : \mathbf{Mod}_S \rightarrow \mathbf{Mod}_R$ (resp. $(-)_R = \mathrm{Res}_{S/R} : \mathbf{Alg}_S \rightarrow \mathbf{Alg}_R$), which maps the morphisms to themselves forgetfully.

§1.4 Classical Groups

1.4.1 Convention. Let $X = (x_{i,j})_{1 \leq i, j \leq n}$ be a n -by- n matrix of indeterminants. For any ring R , let

$$R[X] := R[x_{1,1}, x_{1,2}, \dots, x_{n,n}]$$

Abusing notation, identify $X = (x_{i,j})_{1 \leq i, j \leq n} \in M_n(R[X])$. For any $F = (f_{i,j})_{1 \leq i, j \leq n} \in M_n(R[X])$, denote $\det F := \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^n f_{i, \sigma(i)} \in R[X]$ and $\mathrm{tr}(A) := \sum_{i=1}^n f_{i,i} \in R[X]$. Moreover, we denote the ideal $(F) := (f_{1,1}, f_{1,2}, \dots, f_{n,n}) \subseteq R[X]$.

1.4.2 Example. Here are some more examples of algebraic groups over R ,

- (i) the *general linear group* of a free module V over R of rank n ,

$$\mathrm{GL}_V : \mathbf{Alg}_R \rightarrow \mathbf{Grp} \quad S \mapsto \mathrm{Aut}_{\mathbf{Mod}_S}(V_S) = \mathrm{Aut}_{\mathbf{Mod}_S}(V \otimes_R S)$$

where $\widetilde{\mathrm{GL}}_V \cong \mathrm{Spec} \left(\frac{R[X][y]}{(y \det(X) - 1)} \right)$ where $X = (x_{i,j})_{1 \leq i,j \leq n}$ is an indeterminate matrix,

(ii) the *special linear group* of a free module V over R of rank n ,

$$\mathrm{SL}_V : \mathrm{Alg}_R \rightarrow \mathrm{Grp} \quad S \mapsto \mathrm{Ker} \left(\mathrm{GL}_V(S) \xrightarrow{\det_S} S^\times \right)$$

where $\widetilde{\mathrm{SL}}_V \cong \mathrm{Spec} \left(\frac{R[X]}{(\det(X) - 1)} \right)$ where $X = (x_{i,j})_{1 \leq i,j \leq n}$ is an indeterminate matrix.

1.4.3 Convention. Let V be a free R -module of rank n . We denote $\mathrm{GL}_n := \mathrm{GL}_V$ and $\mathrm{SL}_n := \mathrm{SL}_V$.

1.4.4 Definition. Let V be a module over an arbitrary ring R . Suppose $\sigma : R \rightarrow R$ is a ring map satisfying $\sigma \circ \sigma = \mathrm{id}$, which we call an *involution* map. A function

$$b : V \times V \rightarrow R$$

is called a *form* on V , and is said to be

- (i) *left linear* if $b(rv + u, w) = rb(v, w) + b(u, w)$ for all $r \in R$ and $u, v, w \in V$,
- (ii) *left σ -linear* if $b(rv + u, w) = \sigma(r)b(v, w) + b(u, w)$ for all $r \in R$ and $u, v, w \in V$,
- (iii) *right linear* if $b(v, rw + u) = rb(v, w) + b(v, u)$ for all $r \in R$ and $u, v, w \in V$,
- (iv) *bilinear* if it is left linear and right linear,
- (v) *σ -sesquilinear* if it is left σ -linear and right linear,
- (vi) *symmetric* if $b(v, w) = b(w, v)$ for all $v, w \in V$,
- (vii) *σ -symmetric* if $b(v, w) = \sigma(b(w, v))$ for all $v, w \in V$,
- (viii) *alternating* if $b(v, v) = 0$ for all $v \in V$,
- (ix) *nondegenerate* if $b(v, w) = 0$ for all $v \in V$ implies $w = 0$,
- (x) *orthogonal* if it is a bilinear, symmetric, and nondegenerate,
- (xi) *symplectic* if it is a bilinear, alternating, and nondegenerate,
- (xii) *σ -Hermitian* if it is σ -sesquilinear, σ -symmetric, and nondegenerate.

1.4.5 Definition. Let $b_1 : V \times V \rightarrow R$ and $b_2 : V \times V \rightarrow R$ be forms on a

module V over an arbitrary ring R . We say b_1 is *equivalent* (or *isometric*) to b_2 if there exists an invertible linear map $P : V \rightarrow V$, which we call an *isometry*, satisfying $b_1(v, w) = b_2(P(v), P(w))$ for all $v, w \in V$.

1.4.6 Example. Let $V = R^n$ be the free module of rank n over a ring R . The *standard orthogonal form* of signature (p, q) on V , where $p, q \in \mathbb{N}$ such that $p + q = n$, is

$$b_{p,q} : V \times V \rightarrow R \quad (x, y) \mapsto \underbrace{x_1y_1 + \cdots + x_py_p}_p - \overbrace{(x_{p+1}y_{p+1} + \cdots + x_ny_n)}^q$$

where $b_{n,0} = \langle \cdot, \cdot \rangle$ is the *standard inner product*.

1.4.7 Example. Recall that a quadratic form on a free module $V = R^n$ of rank n over R is a quadratic homogeneous polynomial map $q : V \rightarrow R$

$$q(x_1, \dots, x_n) = \sum_{1 \leq i, j \leq n} a_{i,j} x_i x_j$$

Recall the polar form $b_q : V \times V \rightarrow R$ of a quadratic form $q : V \rightarrow R$ is the symmetric bilinear form

$$b_q(x, y) = q(x + y) - q(x) - q(y)$$

and we say q is nonsingular if b_q is nondegenerate. Suppose 2 is invertible in R , then we can define the polarization $b = \frac{1}{2}b_q$ of q , which satisfies $q(x) = b(x, x)$. Therefore, when 2 is invertible in R , a quadratic form corresponds uniquely to a symmetric bilinear form.

1.4.8 Example. Let $V = R^{2n}$ be the free module of rank $2n$ over a ring R , the form $\omega : V \times V \rightarrow R$ given by $\omega(x, y) = x^T J y$, where $J \in M_{2n}(R)$ is the *standard symplectic matrix*

$$J := \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

with I the n -by- n identity matrix, is a symplectic form called the *standard symplectic form*. This is in fact the unique symplectic form in dimension $2n$ up to isometry if R is a field. In odd dimensions, there are no symplectic forms at all.

1.4.9 Definition. A *quadratic étale algebra* E over a ring R is a R -algebra such that

$$E \cong R[\omega]/(\omega^2 + b\omega + c)$$

for some $b, c \in R$ where $b^2 - 4c \in R^\times$. Define its *involution* as the unique

nontrivial automorphism

$$\sigma : E \rightarrow E \quad \omega \mapsto -b - \omega$$

which satisfies $\sigma \circ \sigma = \text{id}_E$ and fixes R .

1.4.10 Example. Here are some examples of quadratic étale algebras

- (i) $\mathbb{C}$ as a $\mathbb{R}$ -algebra, with $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ given by $\sigma(z) = \bar{z}$
- (ii) any quadratic extension $K(\sqrt{d})$ of a field K , with $\sigma : K(\sqrt{d}) \rightarrow K(\sqrt{d})$ by $\sqrt{d} \mapsto -\sqrt{d}$
- (iii) $\mathbb{F}_{q^2}$ as a $\mathbb{F}_q$ -algebra, with $\sigma : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{q^2}$ given by $\sigma(x) = x^q$

1.4.11 Example. Let E be a quadratic étale algebra over R with involution $\sigma : E \rightarrow E$. The *standard σ -Hermitian form* of signature (p, q) on $V = E^n$, where $p, q \in \mathbb{N}$ such that $p + q = n$, is

$$h_{p,q} : V \times V \rightarrow E \quad (x, y) \mapsto \underbrace{\sigma(x_1)y_1 + \cdots + \sigma(x_p)y_p}_p - \overbrace{(\sigma(x_{p+1})y_{p+1} + \cdots + \sigma(x_n)y_n)}^q$$

where $h_{n,0}$ is just known as the *standard σ -Hermitian form*.

1.4.12 Definition. Let $V = R^n$ be the free module of rank n over a ring R with an ordered basis $e = (e_1, \dots, e_n)$. Let $b : V \times V \rightarrow R$ be a form, then the *Gram matrix* of b with respect to ordered basis e , is the matrix $[b]_e := (b(e_i, e_j))_{1 \leq i, j \leq n} \in M_n(R)$.

1.4.13 Proposition. Let $V = R^n$ be a free R -module of rank n , let e be an ordered basis of V , and let $b : V \times V \rightarrow R$ be an orthogonal form, then

- (i) $b(x, y) = [x]_e^T [b]_e [y]_e$ for all $x, y \in V$,
- (ii) $[b]_e^T = [b]_e$,
- (iii) $[b]_e$ is invertible,
- (iv) $[b]_{Pe} = [P]_e^T [b]_e [P]_e$ for any invertible linear map $P : V \rightarrow V$.

Proof. Left as exercise. □

1.4.14 Definition. Let R be an arbitrary ring, suppose S is a R -algebra, V is a module over R , and $b : V \times V \rightarrow R$ a bilinear form on V . The *base change* of b from R to S is the bilinear form

$$b_S : V_S \times V_S \rightarrow S \quad (v \otimes r, w \otimes s) \mapsto b(v, w)rs$$

on $V_S = V \otimes_R S$, extended bilinearly.

1.4.15 Example. Let V be a free R -module of rank n with an orthogonal form $b : V \times V \rightarrow R$. Here are some more examples of algebraic groups over R .

- (i) the *orthogonal group* of the pair (V, b)

$$\mathrm{O}_{V,b} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{O}_{V,b}(S) = \{g \in \mathrm{Aut}_{\mathrm{Mod}_S}(V_S) : \forall v, w \in V_S, b_S(gv, gw) = b_S(v, w)\}$$

where for an indeterminant matrix $X = (x_{i,j})_{1 \leq i, j \leq n}$

$$\widetilde{\mathrm{O}_{V,b}} = \mathrm{Spec} \left(\frac{R[X]}{(X^T[b]_e X - [b]_e)} \right)$$

where e is an ordered basis of V , on which $\mathrm{O}_{V,b}$ does not depend,

- (ii) the *special orthogonal group* of the pair (V, b)

$$\mathrm{SO}_{V,b} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{SO}_{V,b}(S) = \{g \in \mathrm{O}_{V,b}(S) : \det(g) = 1\}$$

where for an indeterminant matrix $X = (x_{i,j})_{1 \leq i, j \leq n}$

$$\widetilde{\mathrm{SO}_{V,b}} = \mathrm{Spec} \left(\frac{R[X]}{(X^T[b]_e X - [b]_e, \det X - 1)} \right)$$

where e is an ordered basis of V , on which $\mathrm{SO}_{V,b}$ does not depend,

- (iii) the *orthogonal semilitude group* of the pair (V, b)

$$\mathrm{GO}_{V,b} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{GO}_{V,b}(S) = \{g \in \mathrm{Aut}_{\mathrm{Mod}_S}(V_S) : \exists \lambda \in S^\times, \forall v, w \in V_S, b_S(gv, gw) = \lambda b_S(v, w)\}$$

where for an indeterminant matrix $X = (x_{i,j})_{1 \leq i, j \leq n}$

$$\widetilde{\mathrm{GO}_{V,b}} = \mathrm{Spec} \left(\frac{R[X][\lambda, \nu]}{(X^T[b]_e X - \lambda[b]_e, \lambda\nu - 1)} \right)$$

where e is an ordered basis of V , on which $\mathrm{GO}_{V,b}$ does not depend,

- (iv) the *special orthogonal semilitude group* of the pair (V, b)

$$\mathrm{GSO}_{V,b} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{GSO}_{V,b}(S) = \{g \in \mathrm{GO}_{V,b}(S) : \det(g) = 1\}$$

where for an indeterminant matrix $X = (x_{i,j})_{1 \leq i,j \leq n}$

$$\widetilde{\mathrm{GSO}}_{V,b} = \mathrm{Spec} \left(\frac{R[X][\lambda, \nu]}{(X^T[b]_e X - \lambda[b]_e, \det(X) - 1, \lambda\nu - 1)} \right)$$

where e is an ordered basis of V , on which $\mathrm{GSO}_{V,b}$ does not depend,

1.4.16 Remark. Let $V = R^n$ be the free R -module of rank n and $b = \langle \cdot, \cdot \rangle$ the standard inner product, then we denote $\mathrm{O}_n := \mathrm{O}_{V,b}$, $\mathrm{SO}_n := \mathrm{SO}_{V,b}$, $\mathrm{GO}_n := \mathrm{GO}_{V,b}$, and $\mathrm{GSO}_n := \mathrm{GSO}_{V,b}$.

1.4.17 Proposition. Let $V = R^{2n}$ be a free R -module of rank $2n$, let e be an ordered basis of V , and let $\omega : V \times V \rightarrow R$ be an symplectic form, then

- (i) $\omega(x, y) = [x]_e^T [\omega]_e [y]_e$ for all $x, y \in V$
- (ii) $[\omega]_e^T = -[\omega]_e$ and $[\omega]_e$ has zero diagonal,
- (iii) $[\omega]_e$ is invertible,
- (iv) $[\omega]_{Pe} = [P]_e^T [\omega]_e [P]_e$ for any invertible linear map $P : V \rightarrow V$.
- (v) if R is a field then there exists ordered basis b such that $[\omega]_b = J$ where

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

is the standard symplectic matrix.

Proof. Left as exercise. □

1.4.18 Example. Let V be a free R -module of rank $2n$ with an symplectic form $\omega : V \times V \rightarrow R$. Here are some more examples of algebraic groups over R .

- (i) the *symplectic group* of the pair (V, ω)

$$\mathrm{Sp}_{V,\omega} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{Sp}_{V,\omega}(S) = \{g \in \mathrm{Aut}_{\mathrm{Mod}_S}(V_S) : \forall v, w \in V_S, \omega_S(gv, gw) = \omega_S(v, w)\}$$

where for an indeterminant matrix $X = (x_{i,j})_{1 \leq i,j \leq n}$

$$\widetilde{\mathrm{Sp}}_{V,\omega} = \mathrm{Spec} \left(\frac{R[X]}{(X^T[\omega]_e X - [\omega]_e)} \right)$$

where e is an ordered basis for V , on which $\mathrm{Sp}_{V,\omega}$ does not depend.

(ii) the *symplectic semilitude group* of the pair (V, ω)

$$\mathrm{GSp}_{V,\omega} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{GSp}_{V,\omega}(S) = \{g \in \mathrm{Aut}_{\mathrm{Mod}_S}(V_S) : \exists \lambda \in S^\times, \forall v, w \in V_S, \omega_S(gv, gw) = \lambda \omega_S(v, w)\}$$

where for an indeterminant matrix $X = (x_{i,j})_{1 \leq i, j \leq n}$

$$\widetilde{\mathrm{GSp}_{V,\omega}} = \mathrm{Spec} \left(\frac{R[X][\lambda, \nu]}{(X^T[\omega]_e X - \lambda[\omega]_e, \lambda\nu - 1)} \right)$$

where e is an ordered basis for V , on which $\mathrm{Sp}_{V,\omega}$ does not depend.

1.4.19 Remark. Let $V = R^{2n}$ be the free R -module of rank $2n$ and ω the standard symplectic form on V , then we denote $\mathrm{Sp}_{2n} := \mathrm{Sp}_{V,\omega}$ and $\mathrm{GSp}_{2n} := \mathrm{GSp}_{V,\omega}$.

1.4.20 Definition. Let R be an arbitrary ring, and E a quadratic étale R -algebra with involution $\sigma : E \rightarrow E$. Suppose V is a module over E , and $h : V \times V \rightarrow E$ a σ -Hermitian form on V . The *base change* of h from R to a R -algebra S is the σ -Hermitian form

$$h_S : V_{E_S} \times V_{E_S} \rightarrow E_S \quad (v \otimes r, w \otimes s) \mapsto (h(v, w) \otimes_R 1_S) \sigma_S(r) s$$

extended σ -sesquilinearly, on $V_{E_S} = V \otimes_E E_S$ where $E_S = E \otimes_R S$ is the quadratic étale algebra over S with involution $\sigma_S = \sigma \otimes_R \mathrm{id}_S : E_S \rightarrow E_S$.

1.4.21 Proposition. Let E be a quadratic étale R -algebra with involution $\sigma : E \rightarrow E$. Let $V = E^n$ be a free E -module of rank n , let e be an ordered basis of V , and let $h : V \times V \rightarrow E$ be an σ -Hermitian form, then

$$(i) \quad h(x, y) = [x]_e^* [h]_e [y]_e \text{ for all } x, y \in V$$

$$(ii) \quad [h]_e^* = [h]_e,$$

$$(iii) \quad [h]_e \text{ is invertible,}$$

$$(iv) \quad [h]_{Pe} = [P]_e^* [h]_e [P]_e \text{ for any invertible linear map } P : V \rightarrow V.$$

where $M^* := (\sigma(m_{j,i})) \in M_{k \times \ell}(R)$ denotes the σ -conjugate transpose of $M = (m_{i,j}) \in M_{\ell \times k}(R)$.

Proof. Left as exercise. □

1.4.22 Remark. Let E be a quadratic étale R -algebra, then we can write each element of E uniquely as $a + b\omega$ where $a, b \in R$ with ω as in [Definition 1.4.9](#).

Let $X \in M_{n,m}(E[x_1, \dots, x_\ell])$, define $(X)_{1,\omega}$ as the ideal in $R[x_1, \dots, x_\ell]$, generated by the entries of $A, B \in M_n(R[x_1, \dots, x_\ell])$ where $X = A + B\omega$.

1.4.23 Remark. Let E be a quadratic étale R -algebra. Let $X = (x_{i,j})_{1 \leq i,j \leq n}$ be an indeterminant matrix, for $F = (f_{i,j})_{1 \leq i,j \leq n} \in M_n(E[X])$, denote its σ -conjugate transpose $F^* = (\sigma(f_{j,i}))_{1 \leq i,j \leq n}$ where we extend $\sigma : E \rightarrow E$ to $\sigma : E[X] \rightarrow E[X]$ by fixing any formal variables. Similarly, we can define σ -conjugate transpose of $F \in E[X_1, \dots, X_\ell]$ where $X_1, \dots, X_\ell$ are indeterminant matrices.

1.4.24 Example. Let E be a quadratic étale R -algebra with involution $\sigma : E \rightarrow E$, V a free E -module of rank n , and $h : V \times V \rightarrow E$ a σ -Hermitian form. Here are some more examples of algebraic groups over R .

(i) the *unitary group* of the pair (V, h)

$$U_{V,h} : \text{Alg}_R \rightarrow \text{Grp}$$

where for each R -algebra S

$$U_{V,h}(S) = \{g \in \text{Aut}_{\text{Mod}_S}(V_{E_S}) : \forall v, w \in V_{E_S}, h_S(gv, gw) = h_S(v, w)\}$$

where for indeterminant matrices $X = (x_{i,j})_{1 \leq i,j \leq n}$ and $Y = (y_{i,j})_{1 \leq i,j \leq n}$

$$\widetilde{U_{V,h}} = \text{Spec} \left(\frac{R[X, Y][y]}{((X + Y\omega)^*[h]_e(X + Y\omega) - [h]_e, y \det(X + Y\omega) - 1)_{1,\omega}} \right)$$

where e is an ordered basis of V , on which $U_{V,h}$ does not depend,

(ii) the *special unitary group* of the pair (V, h)

$$SU_{V,h} : \text{Alg}_R \rightarrow \text{Grp}$$

where for each R -algebra S

$$SU_{V,h}(S) = \{g \in U_{V,h}(S) : \det(g) = 1\}$$

where for indeterminant matrices $X = (x_{i,j})_{1 \leq i,j \leq n}$ and $Y = (y_{i,j})_{1 \leq i,j \leq n}$

$$\widetilde{SU_{V,h}} = \text{Spec} \left(\frac{R[X, Y]}{((X + Y\omega)^*[h]_e(X + Y\omega) - [h]_e, \det(X + Y\omega) - 1)_{1,\omega}} \right)$$

where e is an ordered basis of V , on which $SU_{V,h}$ does not depend,

(iii) the *unitary semilitude group* of the pair (V, h)

$$GU_{V,h} : \text{Alg}_R \rightarrow \text{Grp}$$

where for each R -algebra S

$$\mathrm{GU}_{V,h}(S) = \{g \in \mathrm{Aut}_{\mathrm{Mod}_S}(V_{E_S}) : \exists \lambda \in S^\times, \forall v, w \in V_{E_S}, h_S(gv, gw) = (1 \otimes \lambda)h_S(v, w)\}$$

where for indeterminant matrices $X = (x_{i,j})_{1 \leq i,j \leq n}$ and $Y = (y_{i,j})_{1 \leq i,j \leq n}$

$$\widetilde{\mathrm{GU}}_{V,h} = \mathrm{Spec} \left(\frac{R[X, Y][y, \lambda, \nu]}{((X + Y\omega)^*[h]_e(X + Y\omega) - \lambda[h]_e, y \det(X + Y\omega) - 1, \lambda\nu - 1)_{1,\omega}} \right)$$

where e is an ordered basis of V , on which $\mathrm{GU}_{V,h}$ does not depend,

(iv) the *special unitary semilitude group* of the pair (V, h)

$$\mathrm{GSU}_{V,h} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where for each R -algebra S

$$\mathrm{GSU}_{V,h}(S) = \{g \in \mathrm{GU}_{V,h}(S) : \det(g) = 1\}$$

where for indeterminant matrices $X = (x_{i,j})_{1 \leq i,j \leq n}$ and $Y = (y_{i,j})_{1 \leq i,j \leq n}$

$$\widetilde{\mathrm{GSU}}_{V,h} = \mathrm{Spec} \left(\frac{R[X, Y][\lambda, \nu]}{((X + Y\omega)^*[h]_e(X + Y\omega) - \lambda[h]_e, \det(X + Y\omega) - 1, \lambda\nu - 1)_{1,\omega}} \right)$$

where e is an ordered basis of V , on which $\mathrm{GSU}_{V,h}$ does not depend,

1.4.25 Remark. The choice of ω for a quadratic étale R -algebra E is not necessarily unique, but one can show that none of the groups in [Example 1.4.24](#) depend on the choice of ω .

1.4.26 Definition. Let V be a free module over R and $q : V \rightarrow R$ a quadratic form. The *Clifford algebra* $\mathrm{Cl}(V, q)$ is the universal associative unital R -algebra generated by V and satisfying $v^2 = q(v)1$, in the sense that for any unital associative R -algebra C and R -linear map $f : V \rightarrow C$ such that $f(v)^2 = q(v)1$, there exists unique R -algebra morphism $\tilde{f} : \mathrm{Cl}(V, q) \rightarrow C$ such that $f = \tilde{f} \circ i$ where $i : V \rightarrow \mathrm{Cl}(V, q)$ is the canonical inclusion. By the universal property, the map $V \rightarrow \mathrm{Cl}(V, q)$ by $v \mapsto -v$ extends to a map $\alpha : \mathrm{Cl}(V, q) \rightarrow \mathrm{Cl}(V, q)$ called the *involution*, and therefore we have a canonical $\mathbb{Z}/2$ -grading on the Clifford algebra $\mathrm{Cl}(V, q) = \mathrm{Cl}^0(V, q) \otimes \mathrm{Cl}^1(V, q)$ where

$$\begin{aligned} \mathrm{Cl}^0(V, q) &= \{x \in \mathrm{Cl}(V, q) : \alpha(x) = x\} \\ \mathrm{Cl}^1(V, q) &= \{x \in \mathrm{Cl}(V, q) : \alpha(x) = -x\} \end{aligned}$$

making the Clifford algebra a $\mathbb{Z}/2$ -graded algebra.

1.4.27 Remark. Let V be a free module over R and $q : V \rightarrow R$ a quadratic

form. Constructively, one can define the Clifford algebra to be

$$\mathrm{Cl}(V, q) = \bigoplus_{i \in \mathbb{N}} R^{\otimes i} / (v \otimes v - q(v)1)$$

from which the $\mathbb{Z}/2$ -grading is inherited.

1.4.28 Definition. Let V be a free module over R and $q : V \rightarrow R$ a quadratic form. Let $\mathrm{Cl}(V, q)^{\mathrm{op}}$ be the opposite algebra of the Clifford algebra, i.e. it has the same underlying set and additive structure but with xy defined to be yx in $\mathrm{Cl}(V, q)$. By the universal property, the map $V \rightarrow \mathrm{Cl}(V, q)^{\mathrm{op}}$ given by $v \mapsto -v$ extends to a map $c : \mathrm{Cl}(V, q) \rightarrow \mathrm{Cl}(V, q)^{\mathrm{op}}$, which after composing with the canonical identification $\mathrm{Cl}(V, q)^{\mathrm{op}} \cong \mathrm{Cl}(V, q)$ gives a function

$$\overline{(\cdot)} : \mathrm{Cl}(V, q) \rightarrow \mathrm{Cl}(V, q) \quad x \mapsto \bar{x}$$

which we call the *Clifford conjugation*.

1.4.29 Remark. Let V be a free module over R and $q : V \rightarrow R$ a quadratic form. Let S be an R -algebra, we write the base change $\mathrm{Cl}_S(V, q) = \mathrm{Cl}(V, q) \otimes_R S \cong \mathrm{Cl}(V_S, q_S)$, where $q_S : V_S \rightarrow S$ is the base change of the quadratic form defined by $(v \otimes s) \mapsto q(v)s^2$.

1.4.30 Example. Let V be a free module over R and $q : V \rightarrow R$ a quadratic form. Here are some more examples of algebraic groups over R .

(i) the *Clifford group* or *similitude pin group* of the pair (V, q)

$$\mathrm{GPin}_{V,q} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where

$$\mathrm{GPin}_{V,q}(S) = \{x \in \mathrm{Cl}_R(V, q)^{\times} : xV_Sx^{-1} \subseteq V_S\}$$

for each R -algebra S ,

(ii) the *pin group* of the pair (V, q)

$$\mathrm{Pin}_{V,q} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where

$$\mathrm{Pin}_{V,q}(S) = \{x \in \mathrm{GPin}_{V,q}(S) : x\bar{x} = 1\}$$

for each R -algebra S ,

(iii) the *similitude spin group* of the pair (V, q)

$$\mathrm{GSpin}_{V,q} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where

$$\mathrm{GSpin}_{V,q}(S) = \mathrm{GPin}_{V,q}(S) \cap \mathrm{Cl}^0(V, q)$$

for each R -algebra S ,

(iv) the *spin group* of the pair (V, q)

$$\mathrm{Spin}_{V,q} : \mathrm{Alg}_R \rightarrow \mathrm{Grp}$$

where

$$\mathrm{Spin}_{V,q}(S) = \{x \in \mathrm{GSpin}_{V,q}(S) : x\bar{x} = 1\}$$

for each R -algebra S .

1.4.31 Remark. Another important family of algebraic groups is the *exceptional groups*: G_2, F_4, E_6, E_7 , and E_8 , which we will not define until later.

1.4.32 Definition. Let G be an algebraic group over R and S a finite projective R -algebra. Define

- (i) *base change functor* $(-)_S : \mathrm{GAlg}_R \rightarrow \mathrm{GAlg}_S$ given by $G_S(T) = G(T)$ for any S -algebra T , and maps morphisms to themselves forgetfully;
- (ii) *Weil restriction functor* $(-)_R : \mathrm{GAlg}_S \rightarrow \mathrm{GAlg}_R$ given by $G_R(T) = G(T \otimes_R S)$ for any R -algebra T , in other words, $G_R = G \circ (- \otimes_R S)$, and for a morphism $\varphi : G \rightarrow H$ of algebraic groups over S , the induced map $\varphi_R : G_R \rightarrow H_R$ is $(\varphi_R)_T = \varphi_{T \otimes_R S}$ for each R -algebra T .

1.4.33 Remark. Recall that we say a R -algebra S is finite projective if there exists a R -module T such that $S \oplus T$ is a free module of finite rank. The assumption of finite projectiveness in [Definition 1.4.32](#) is to ensure the base changed or restricted group functors are indeed algebraic groups.

1.4.34 Example.

dimension connectedness, irreducibility products, semidirect product, identity component, open/closed subgroups, actions, kernel, image, generators

§1.5 Commutative Hopf Algebras

§1.6 Jordan Decomposition

§1.7 Lie Algebra of an Algebraic Group

§1.8 Root Systems and Root Datum

§1.9 Isomorphism and Existence Theorems

Chapter 2

Representations of Linear Algebraic Group

§2.1 Representations of Split Reductive Groups

§2.2 Tannakian Duality

Chapter 3

More on Lie Algebras

Chapter 4

Homogeneous Spaces

§4.1 Toric Varieties

§4.2 Flag Varieties

§4.3 Spherical Varieties

Chapter 5

Invariant Theory

Chapter A

Appendix

Bibliography

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